2MBI75N-120

IGBT Module

1200V / 75A 2 in one-package



■ Features

· High speed switching
· Voltage drive
· Low inductance module structure

■ Applications

· Inverter for Motor drive
· AC and DC Servo drive amplifier
· Uninterruptible power supply
· Industrial machines, such as Welding machines

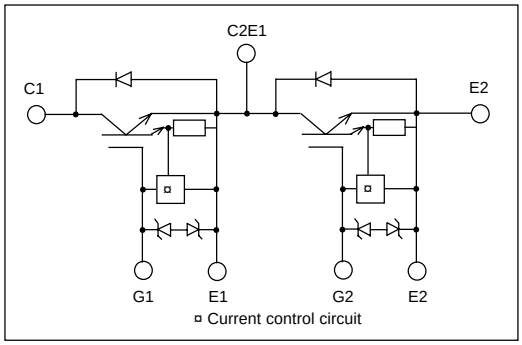
■ Maximum ratings and characteristics

● Absolute maximum ratings (at Tc=25°C unless otherwise specified)

Item		Symbol	Rating	Unit
Collector-Emitter voltage		V_{CES}	1200	V
Gate-Emitter voltaga		V_{GES}	±20	V
Collector current	Continuous	I_C	75	A
	1ms	I_C pulse	150	A
	Continuous	-I_C	75	A
	1ms	-I_C pulse	150	A
Max. power dissipation		P_C	600	W
Operating temperature		T_j	+150	°C
Storage temperature		T_{stg}	-40 to +125	°C
Isolation voltage		V_{is}	AC 2500 (1min.)	V
Screw torque		Mounting *1	3.5	N·m
		Terminals *1	3.5	N·m

*1 : Recommendable value : 2.5 to 3.5 N·m(M5)

■ Equivalent Circuit Schematic



● Electrical characteristics (at Tj=25°C unless otherwise specified)

Item	Symbol	Characteristics			Conditions	Unit
		Min.	Typ.	Max.		
Zero gate voltage collector current	I_{CES}	–	–	1.0	V_{GE}=0V, V_{CE}=1200V	mA
Gate-Emitter leakage current	I_{GES}	–	–	15	V_{CE}=0V, V_{GE}=±20V	µA
Gate-Emitter threshold voltage	$V_{GE(th)}$	4.5	–	7.5	V_{CE}=20V, I_C=75mA	V
Collector-Emitter saturation voltage	$V_{CE(sat)}$	–	–	3.3	V_{GE}=15V, I_C=75A	V
Input capacitance	C_{ies}	–	12000	–	V_{GE}=0V	pF
Output capacitance	C_{oes}	–	4350	–	V_{CE}=10V	
Reverse transfer capacitance	C_{res}	–	3870	–	f=1MHz	
Turn-on time	t_{on}	–	0.65	1.2	V_{CC}=600V	µs
	t_r	–	0.25	0.6	I_C=75A	
Turn-off time	t_{off}	–	0.85	1.5	V_{GE}=±15V	
	t_f	–	0.35	0.5	R_G=16 ohm	
Diode forward on voltage	V_F	–	–	3.0	I_F=75A, V_{GE}=0V	V
Reverse recovery time	t_{rr}	–	–	0.35	I_F=75A	µs

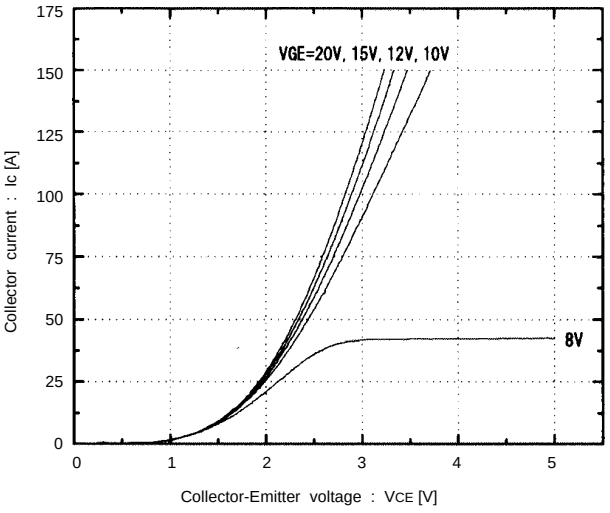
● Thermal resistance characteristics

Item	Symbol	Characteristics			Conditions	Unit
		Min.	Typ.	Max.		
Thermal resistance	Rth(j-c)	–	–	0.21	IGBT	°C/W
	Rth(j-c)	–	–	0.47	Diode	°C/W
	Rth(c-f)*2	–	0.05	–	the base to cooling fin	°C/W

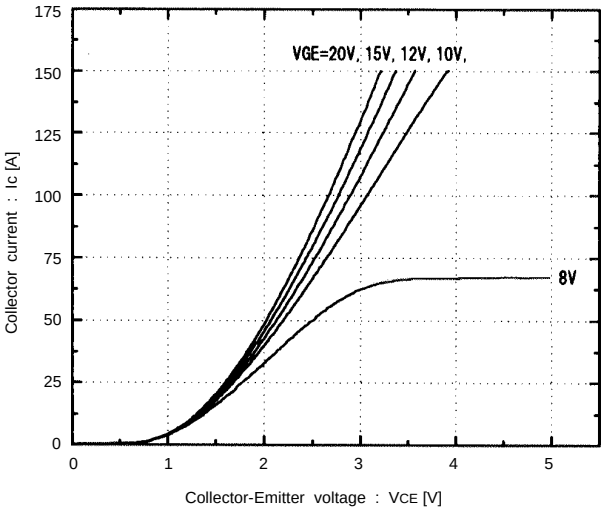
*2 : This is the value which is defined mounting on the additional cooling fin with thermal compound

■ Characteristics (Representative)

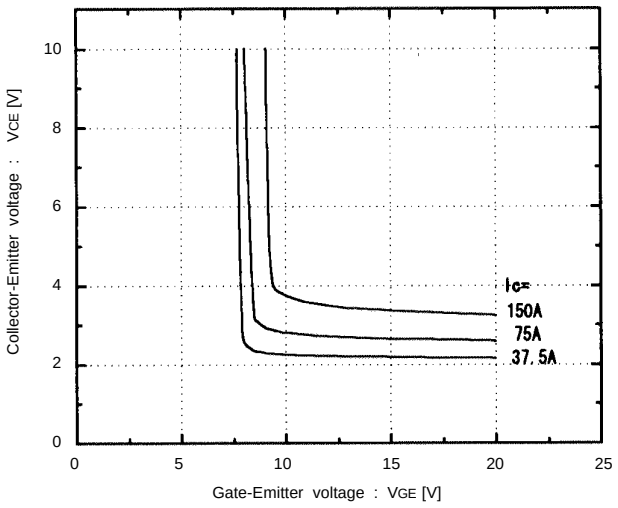
Collector current vs. Collector-Emitter voltage
Tj=25°C



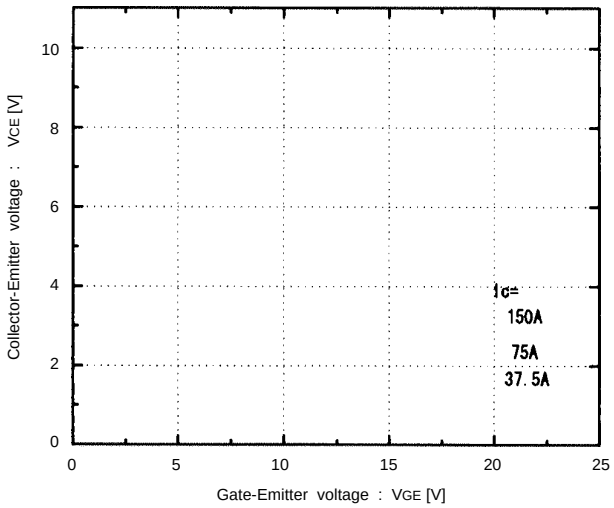
Collector current vs. Collector-Emitter voltage
Tj=125°C



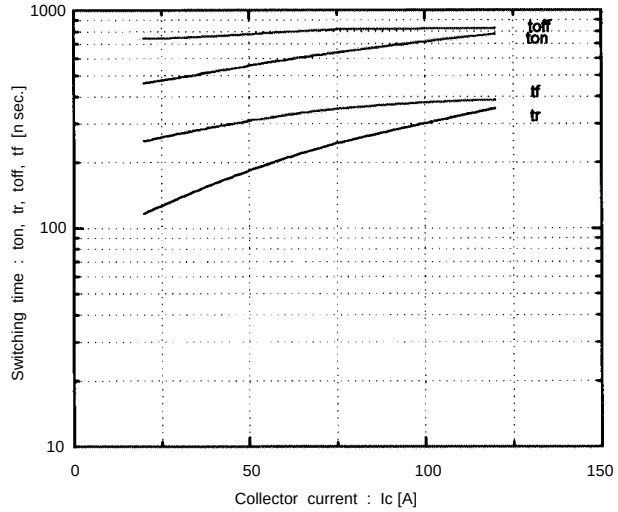
Collector-Emitter vs. Gate-Emitter voltage
Tj=25°C



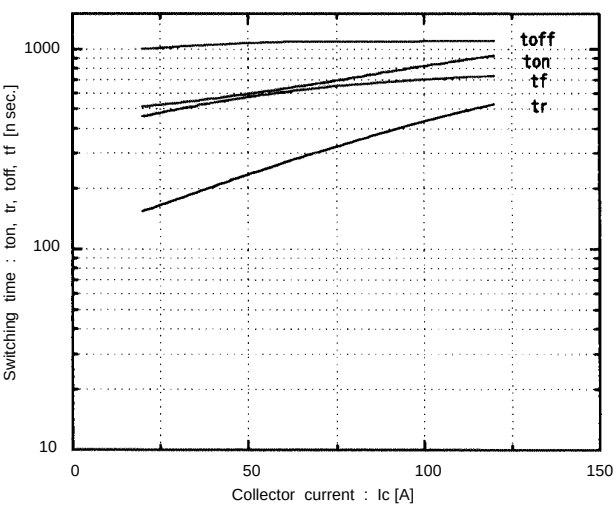
Collector-Emitter vs. Gate-Emitter voltage
Tj=125°C



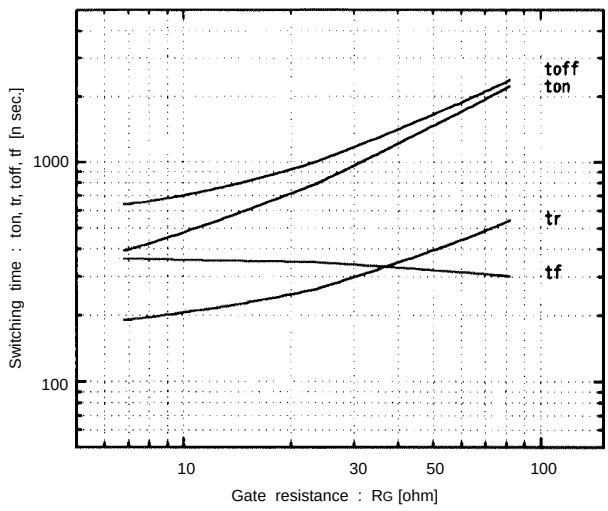
Switching time vs. Collector current
Vcc=600V, RG=16 ohm, VGE=±15V, Tj=25°C



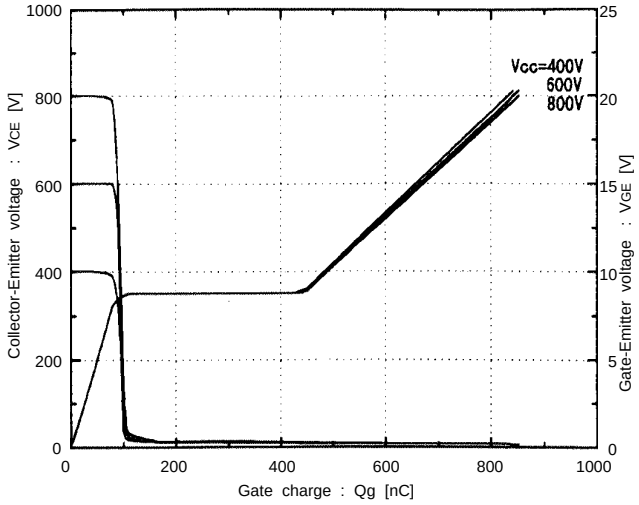
Switching time vs. Collector current
Vcc=600V, RG=16 ohm, VGE=±15V, Tj=125°C



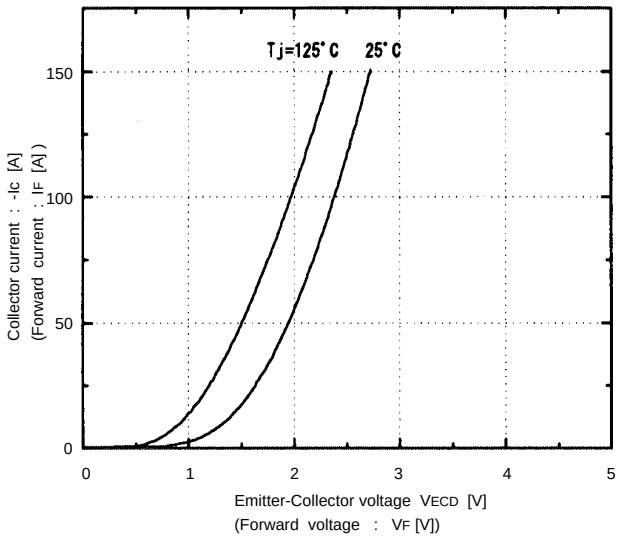
Switching time vs. RG
Vcc=600V, Ic=75A, VGE=±15V, Tj=25°C



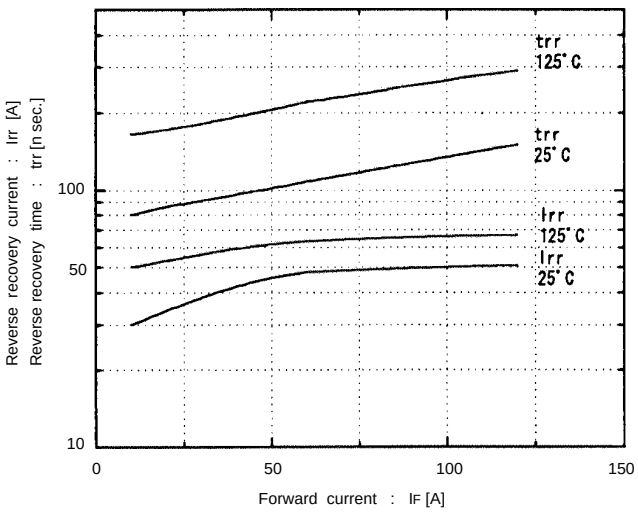
Dynamic input characteristics
Tj=25°C



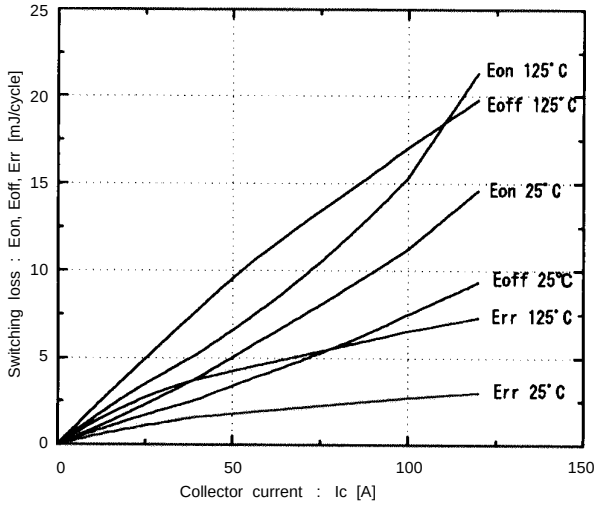
Forward current vs. Forward voltage
VGE=0V



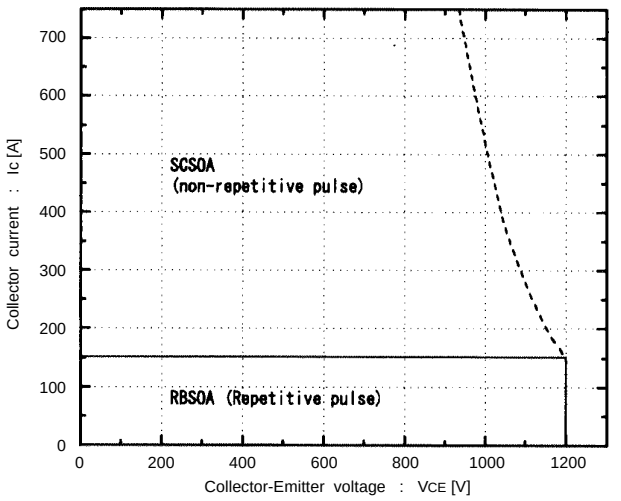
Reverse recovery characteristics
trr, Irr, vs. IF



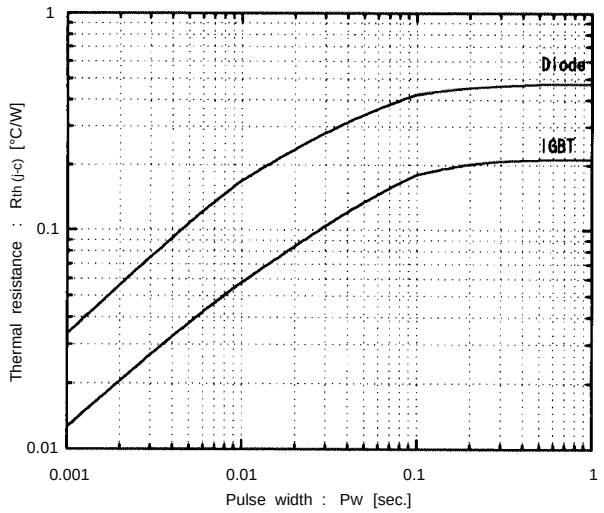
Switching loss vs. Collector current
Vcc=600V, RG=16 ohm, VGE=±15V



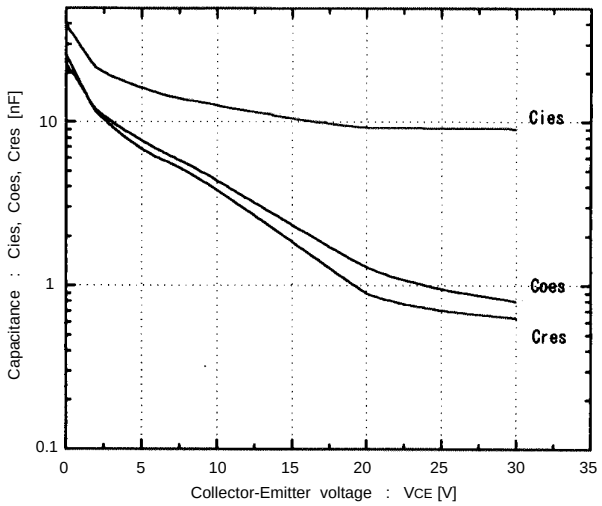
Reversed biased safe operating area
+VGE=15V, -VGE ≦ 15V, Tj ≦ 125°C, RG ≧ 16 ohm



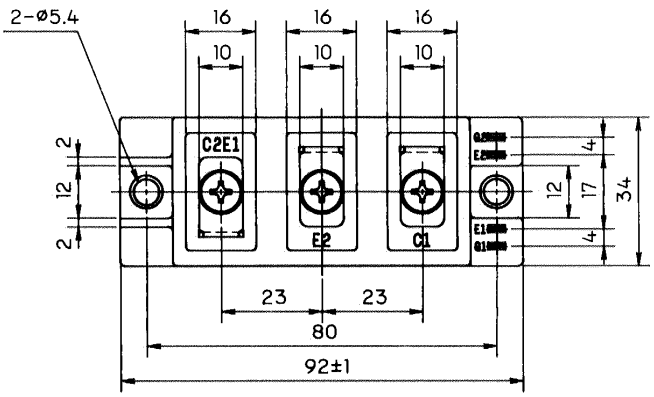
Transient thermal resistance

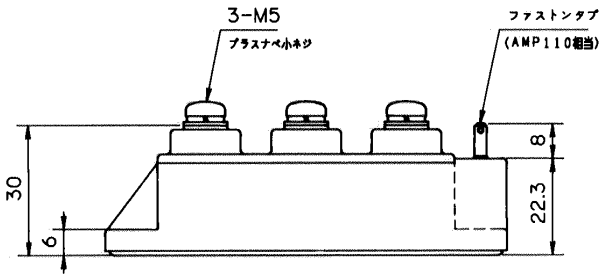


Capacitance vs. Collector-Emitter voltage
Tj=25°C



■ Outline Drawings, mm





mass : 180g